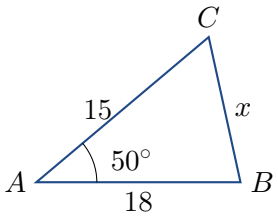
Law of Sines and Law of Cosines

1. Which of the cases listed can lead to more than one solution when solving a triangle (the ambiguous case)?

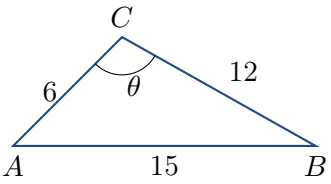
ASA SSA SAS SSS

2. Use the Law of Cosines to determine the indicated side x or angle θ.

A.



B.

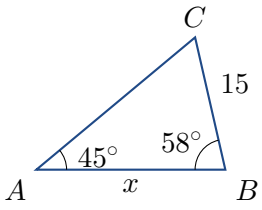


3. Solving a Triangle Solve triangle ABC given $b = 60$, $c = 30$, $\angle A = 70°$.

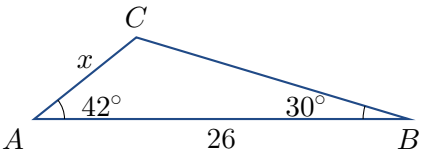
4. Solving a Triangle Solve triangle ABC given $a = 10$, $b = 12$, $c = 16$.

5. Use the Law of Sines to find the indicated side x.

A.



B.



6. Solving a Triangle Use the Law of Sines to solve for all possible triangles that satisfy the given conditions. There may be two triangles, one triangle, or no triangles that satisfy the given conditions.

A. $a = 30$, $c = 40$, $\angle A = 37°$

B. $a = 75$, $b = 100$, $\angle A = 30°$

7. Solving a Triangle Use the Law of Sines to solve for all possible triangles: $b = 73$, $c = 82$, $\angle B = 58°$, $\angle C < 90°$.

8. Mixed Problem In triangle ABC, $\angle A = 40°$, $\angle B = 60°$, and $c = 20$.

A. Use the Law of Sines to find side a.

B. Then use the Law of Cosines to find side b.